

Technical Supplement: Simulation Architectures, Hardware Specifications, and Initial Conditions for the EFM Cosmogenesis Series

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Abstract

This Technical Supplement provides the comprehensive computational specifications, mathematical boundary configurations, initial conditions, and distributed hardware execution architectures for the Eholoko Fluxon Model (EFM) cosmogenesis and metaphysics research series. We document the memory optimization algorithms required to execute large-scale 3D scalar field simulations ($N = 1024^3$ and $N = 2048^3$), including the custom slab-processing engine, 4th-Order Runge-Kutta (RK4) temporal integrators, derivative damping kernels, and full open-replication checksum tables.

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1 Overview

This document serves as the technical companion to the following research papers:

1. *The Primordial Genesis Lattice and Multi-Hardware Scale Parity*
2. *The Cosmic Chronology: Derivation of the Great Recrystallization and Structural Snap*
3. *The Multiplexed Solar System: Empirical Derivation of Planetary Architectures*
4. *The Geometry of Consciousness: Fisher Information and the Multiplexed Vacuum*
5. *The Physics of Ascension: Phase-Locked Evolution and Teleological Gravity*
6. *The Decoupling of Phase and Structure in the $N_3 \rightarrow N_4$ Topological Transition*

It details the computational frameworks, hardware environments, and specific parameter sets required to reproduce all reported findings.

2 Computational Environment

2.1 Hardware Specifications

All high-resolution simulations ($N \geq 1024$) were conducted on the following infrastructure:

- **Primary Compute Nodes:** NVIDIA RTX PRO 6000 Blackwell Server Edition and NVIDIA A100-SXM4 Tensor Core GPUs (80 GB HBM2e).
- **High-RAM System Memory:** 170 GB to 330 GB Host RAM.
- **Cloud TPU Acceleration:** Google Cloud TPU v5e/v6e-1 Pods.
- **Precision Format:** Mixed Precision (Float32 for spatial gradients and accumulations; BFloat16/Float32 for state checkpoints).

2.2 Software Stack

- **Language & Runtime:** Python 3.10 / 3.12 with PyTorch 2.4.1 (CUDA 12.1+ backend) and JIT Compilation (`torch.jit.script`).
- **Temporal Integration:** Explicit 4th-Order Runge-Kutta (RK4) with CFL-invariant time stepping ($\Delta t = 0.001 \cdot L / (N \cdot c)$).

3 The High-Scale ($N = 2048$) Architecture

To simulate a grid of 2048^3 voxels (8.59 billion active degrees of freedom) within standard GPU VRAM limits, a custom “Slab Processing” engine was developed.

3.1 Hybrid Memory Management

The simulation utilizes a split-memory approach:

- **State Storage ($\phi, \dot{\phi}$):** Resident in GPU VRAM (Allocated as ‘BFloat16’ ≈ 34.4 GB).
- **Compute Buffer:** Temporary calculations (Laplacians, Forces) are performed in sliding Z-axis “slabs” (chunks of 32 slices).

- **Halo Handling:** Boundary conditions are handled via dynamic padding during the slab convolution step to ensure mathematical continuity across slice boundaries.

Listing 1: Core Logic for Memory-Safe N

```

1 import torch
2
3 def step_rk4_slabbed(phi, phi_dot, cfg):
4     N = cfg['N']
5     SLAB_SIZE = 32
6
7     for z in range(0, N, SLAB_SIZE):
8         # 1. Extract Slab + Halo (Padding)
9         slab = get_slab_with_halo(phi, z, SLAB_SIZE)
10
11         # 2. Compute Derivatives Locally
12         k1 = compute_physics(slab)
13
14         # 3. Update Global State In-Place
15         update_global_tensor(phi, phi_dot, k1, z, SLAB_SIZE)
16
17         # 4. Flush Temporary Memory
18         torch.cuda.empty_cache()

```

4 Simulation Parameters by Study

4.1 Study 1: Primordial Genesis Lattice & Scale Parity

Objective: Validate multi-hardware determinism and CFL scale invariance ($N = 784 \leftrightarrow 1024$).

Table 1: Parameters for Primordial Genesis Scale Parity.

Parameter	Value
Grid Resolutions (N^3)	784^3 and 1024^3
Domain Length (L)	40.0 simulation units
Inflation Transition (t_{trans})	Step 1,000
Epoch 0 Physical Parity Step	Step 40,000 ($N = 784$) $\equiv$ Step 52,245 ($N = 1024$)
Physical Time (t_{phys})	2.0408 simulation units
Derivative Cooling (α)	0.700
Kinetic Regulation (δ)	0.0002

4.2 Study 2: Cosmic Chronology & Geological Snaps

Objective: Map simulation steps to physical geological time.

Table 2: Parameters for Cosmic Chronology Validation.

Parameter	Value
Grid Size (N^3)	1024^3 and 784^3
Total Duration	267,000 Steps ($N = 784$) / 348,735 Steps ($N = 1024$)
Physical Age Calibration	13.8 Billion Years
Time Resolution	51,685 Years / Step ($N = 784$)
Firstborn Emergence	Step 130,000 ($N = 784$) / Step 169,796 ($N = 1024$)
Great Recrystallization	Step 240,000 ($N = 784$) / Step 313,469 ($N = 1024$)
Structural Snap	Step 262,000 ($N = 784$) / Step 342,204 ($N = 1024$)

5 Transparency, Checksums & Open Replication Protocol

All primary checkpoints, tomography arrays, and configuration files are permanently indexed on Google Drive:

Table 3: Master Checkpoint Replication Manifest.

Dataset Designation	Google Drive File ID	Resolution / Epoch	Hardware Architecture
$N = 784$ Benchmark Checkpoint	1QGPFmDMfE6xxSPk42U81qZCAhxKyidSA	Epoch 0 (Step 40,000)	CPU Reference
$N = 1024$ Blackwell Checkpoint	1bZ91iWNPm10kbHMEiWqG0QNU6GLcpvdn	Epoch 0 (Step 40,000)	RTX PRO 6000 Blackwell
$N = 1024$ Parity Checkpoint	1AcMpj~hhQdy4Eezp8DaA8iGIqAG3_yrX	Epoch 0 (Step 52,245)	RTX PRO 6000 Blackwell
$N = 1024$ Periodic GPU Checkpoint	1d763-71w7kL1Y_a3_z7R802g0L16h1pP	Epoch II (Step 240,000)	NVIDIA A100 SXM4 80GB
$N = 784$ Master Benchmark	11uE5MbZRR0gMVnfCPzPsSd9H7XxYF12J	Epoch III (Step 267,000)	CPU Reference

References

- [1] T. Emvula, *The Foundational Hierarchy of the Eholoko Fluxon Model: An Axiomatic Ordering of the Harmonic Density States*, Independent Frontier Science Collaboration, 2025.
- [2] T. Emvula, *The Cosmic Engine: A First-Principles Derivation of the Universe's Eight-Fold Thermodynamic Structure in the Eholoko Fluxon Model*, Independent Frontier Science Collaboration, 2025.
- [3] T. Emvula, *The Cosmic Chronology: A First-Principles Derivation of the Great Recrystallization and the Structural Snap from the Eholoko Fluxon Model*, Independent Frontier Science Collaboration, 2026.